Average Rate of Change

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.03

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

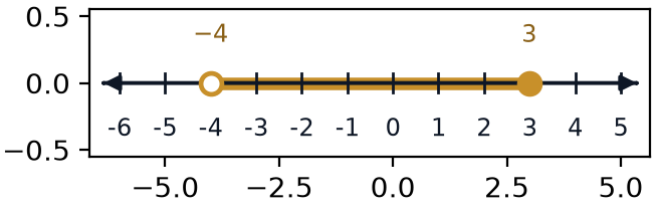
1 [Foundational] • Target: Computing average rate of change

Find the average rate of change of $f(x) = x^2$ over each interval:

a $[0, 2]$ **b** $[1, 3]$ **c** $[2, 5]$ **d** $[-1, 1]$

2 [Foundational] • Target: Computing average rate of change

Find the average rate of change of $f(x) = 2^x$ over the interval $[0, 3]$.



3 [Foundational] • Target: Computing average rate of change

Show that the average rate of change of $g(x) = 3x - 5$ is the same over any interval $[a, b]$, and state its value.

4 [Foundational] • Target: Applications

A ball is thrown upward so that its height after t seconds is $h(t) = -5t^2 + 20t$ meters.

a Find the average rate of change of the height over $[0, 2]$.

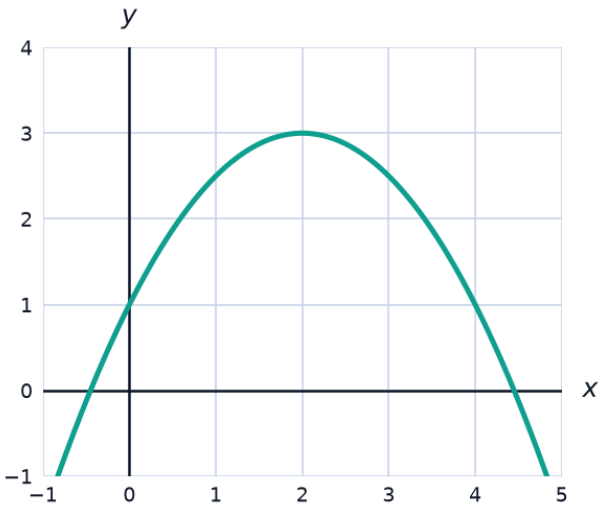
b Find the average rate of change of the height over $[1, 3]$.

c Interpret your answer to part b in the context of the ball's motion.

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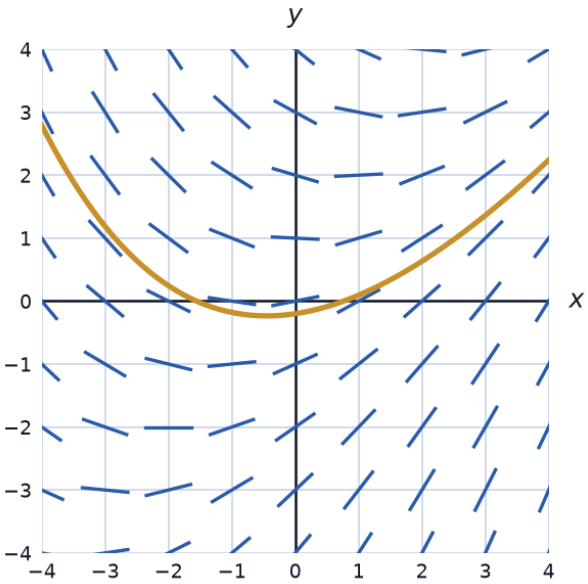
5 [Foundational] • Target: Applications

The graph of $f(x) = -\frac{1}{2}x^2 + 2x + 1$ is shown. Estimate the average rate of change over $[0, 4]$, then verify your estimate algebraically.



6 [Foundational] • Target: Applications

A company's revenue (in thousands of riyals) is modeled by $R(t) = 40 + 6t - 0.5t^2$, where t is years since 2020.



a Find the average rate of change of revenue over $[0, 4]$.

b Find the average rate of change of revenue over $[4, 8]$.

c Compare your two answers and interpret what they say about the company's growth.

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7 [Foundational] • Target: Comparing rates and estimating from a table

A table of values for $g(x)$ is given: $g(1) = 5$, $g(3) = 17$, $g(6) = 44$, $g(10) = 92$. Find the average rate of change of g over:

a $[1, 3]$ **b** $[3, 6]$ **c** $[6, 10]$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Comparing rates and estimating from a table

Based on the previous question, is the average rate of change of g increasing, decreasing, or constant as x increases? What does this suggest about the shape of g's graph?

9 [Intermediate] • Target: Comparing rates and estimating from a table

Two functions model distance traveled: $d_1(t) = 5t$ and $d_2(t) = t^2$ (in km, t in hours). Find the average speed (average rate of change) of each over $[2, 6]$, and state which traveler was faster on average.

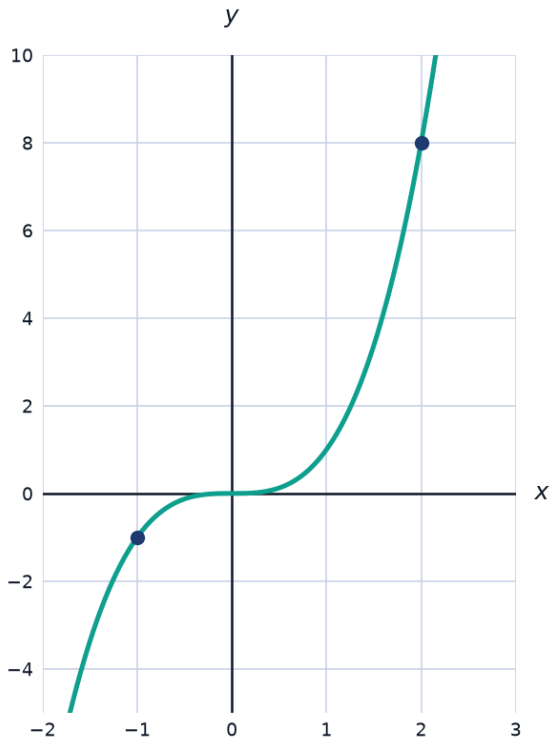
10 [Intermediate] • Target: Average rate of change and secant lines

Explain the geometric meaning of the average rate of change of f over $[a, b]$ in terms of the secant line through $(a, f(a))$ and $(b, f(b))$.

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11 **[Intermediate]** • **Target: Average rate of change and secant lines**

For $f(x) = x^3$, find the average rate of change over $[-1, 2]$, and write the equation of the secant line through the corresponding points.



12 **[Intermediate]** • **Target: Comparing average rate of change across different functions**

Find the average rate of change of $f(x) = 2^x$ and $g(x) = x^2$ over $[0, 4]$, and determine which function grew faster on average.

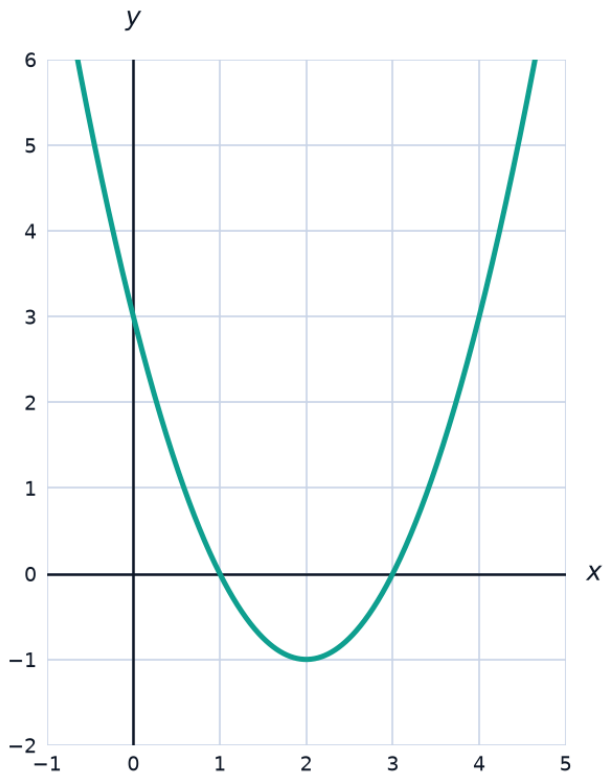
13 **[Intermediate]** • **Target: Comparing average rate of change across different functions**

Find the average rate of change of $f(x) = 2^x$ over $[4, 8]$ and compare it to its value over $[0, 4]$ (found above). What does this reveal about exponential growth?

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14 **[Intermediate]** • **Target: Average rate of change from a graph**

Using the graph of $f(x) = x^2 - 4x + 3$ shown, estimate the average rate of change over $[0, 4]$ by reading endpoint values from the graph, then confirm algebraically.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Units and interpretation in context** • **Cross-Lesson: 6.01 Polar Coordinates and Conversions**

A tank's water volume is $V(t) = 200 - 3t^2$ liters, where t is minutes. Find the average rate of change over $[0, 5]$, including correct units, and interpret its sign.

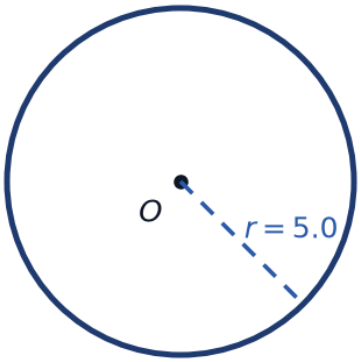
16 **[Challenging]** • **Target: Units and interpretation in context** • **Cross-Lesson: 6.02 Polar Equations and Graphs**

A population model gives $P(t) = 500e^{0.1t}$ (people, years). Find the average rate of change over $[0, 10]$ and state its units, rounding to the nearest whole number.

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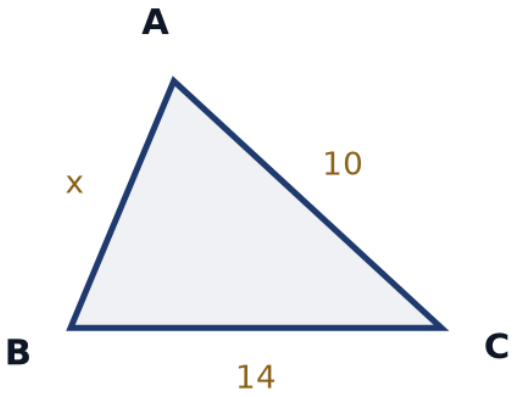
17 [Challenging] • Target: Average rate of change from a graph • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

A circle is centered at the origin and passes through the point $P(a, 4)$ in the first quadrant. The equation of a tangent line to the circle at point P is given by $3x + 4y = k$. Find the value of k.



18 [Challenging] • Target: Units and interpretation in context • Cross-Lesson: 7.02 Logarithms and Their Properties

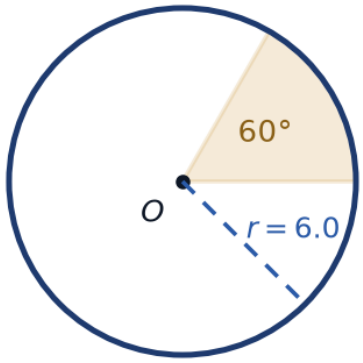
In triangle ABC, angle A is 52 degrees. The side opposite angle B (AC) has a length of 10 units, and the side opposite angle C (AB) has a length of 14 units. What is the length of the side opposite angle A (BC)?



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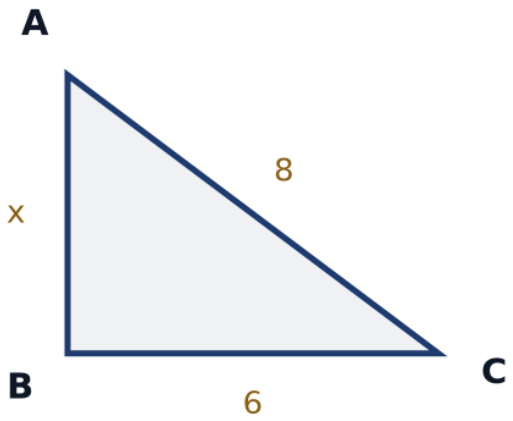
19 [Challenging] • Target: Formula and Secant Line Concept • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

A circle has a radius of 6 units. A chord within this circle subtends a sector with a central angle of 60 degrees. What is the length of the chord?



20 [Challenging] • Target: Tabular and Data-Based Rates of Change • Cross-Lesson: 7.04 Applications: Growth and Decay

In a right-angled triangle *ABC*, the angle *B* is 90 degrees. If the length of side *AB* is 8 units and the length of side *BC* is 6 units, what is the length of the hypotenuse *AC*?



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Formula and Secant Line Concept

State the formula for the average rate of change of $f(x)$ over $[a, b]$.

a. $\frac{f(b) - f(a)}{b - a}$

b. $\frac{f(a) + f(b)}{2}$

c. $\frac{f(b) - f(a)}{2}$

d. $\frac{b - a}{f(b) - f(a)}$

22 [Foundational] • Target: Formula and Secant Line Concept

Find the average rate of change of $f(x) = x^2 - 3x$ from $x = 1$ to $x = 4$.

a. 2

b. 4

c. 3

d. 1

23 [Foundational] • Target: Formula and Secant Line Concept

Find the slope of the secant line connecting $(2, f(2))$ and $(5, f(5))$ for $f(x) = 2x^2 + 1$.

a. 14

b. 12

c. 16

d. 10

24 [Foundational] • Target: Tabular and Data-Based Rates of Change

A table shows $t = [0, 2, 5]$ and distance $s(t) = [0, 24, 75]$. What is the average velocity from $t = 2$ to $t = 5$?

a. 17 m/s

b. 15 m/s

c. 25 m/s

d. 12 m/s

25 [Foundational] • Target: Tabular and Data-Based Rates of Change

Water depth in a reservoir drops from 80 ft at day 4 to 68 ft at day 10. Find the average rate of change in depth per day.

a. -2 ft/day

b. 2 ft/day

c. -1.2 ft/day

d. -12 ft/day

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26 [Foundational] • Target: Tabular and Data-Based Rates of Change

If the average rate of change of $g(x)$ over $[1, 6]$ is 4 and $g(1) = 5$, find $g(6)$.

a. 25

b. 20

c. 29

d. 24

27 [Foundational] • Target: Algebraic Expressions with Variables

Find the average rate of change of $f(x) = x^2$ over the interval $[2, 2 + h]$ in terms of h.

a. $4 + h$

b. 4

c. $2 + h$

d. $4 + 2h$

28 [Foundational] • Target: Algebraic Expressions with Variables

Find the average rate of change of $f(x) = 3x^2 - 5$ over $[x, x + h]$.

a. $6x + 3h$

b. $6x$

c. $3x + 3h$

d. $6x - 5$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Algebraic Expressions with Variables

For $f(x) = \frac{1}{x}$, find the average rate of change over $[a, b]$.

a. $-\frac{1}{ab}$

b. $\frac{1}{ab}$

c. $-\frac{1}{a+b}$

d. $\frac{a-b}{ab}$

30 [Intermediate] • Target: Contextual Applications

A projectile's height is given by $h(t) = -16t^2 + 80t$. Find its average velocity over the first 3 seconds $[0, 3]$.

a. 32 ft/s

b. 48 ft/s

c. 16 ft/s

d. 80 ft/s

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31 [Intermediate] • Target: Contextual Applications

A company's revenue in thousands of dollars is $R(x) = 50x - x^2$, where x is in hundreds of units. What is the average rate of change in revenue as production increases from $x = 10$ to $x = 20$?

a. 20 thousand dollars per unit

b. 30

c. 15

d. 25

32 [Intermediate] • Target: Contextual Applications

Why does a linear function $f(x) = mx + b$ have the same average rate of change over EVERY interval?

a. Because the slope of the secant line between any two points is constantly m

b. Because $b = 0$

c. Because the function has no intercepts

d. Because $f(x)$ is symmetric

33 [Intermediate] • Target: Applications

For $0 < x < 1$, simplify $\cos\left(\sin^{-1}(x)\right)$ as an algebraic expression in x.

a. $\sqrt{1 - x^2}$

b. $\frac{1}{\sqrt{1 - x^2}}$

c. $1 - x^2$

d. $\sqrt{x^2 - 1}$

34 [Intermediate] • Target: Comparing rates and estimating from a table

For $0 < x < 1$, simplify $\sec\left(\tan^{-1}(x)\right)$ as an algebraic expression in x.

a. $1 + x^2$

b. $\frac{1}{\sqrt{1 + x^2}}$

c. $\sqrt{1 - x^2}$

d. $\sqrt{1 + x^2}$

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35 [Intermediate] • Target: Average rate of change and secant lines

For $0 < x < 1$, simplify $\cos\left(\sin^{-1}\left(\frac{x}{5}\right)\right)$ as an algebraic expression in x.

a. $\sqrt{25 - x^2}$

b. $\frac{\sqrt{25 - x^2}}{5}$

c. $\frac{x}{\sqrt{25 - x^2}}$

d. $\frac{\sqrt{25 + x^2}}{5}$

36 [Intermediate] • Target: Comparing average rate of change across different functions

For $0 < x < 1$, simplify $\cos\left(\sin^{-1}\left(\frac{x}{2}\right)\right)$ as an algebraic expression in x.

a. $\sqrt{4 - x^2}$

b. $\frac{\sqrt{4 + x^2}}{2}$

c. $\frac{x}{\sqrt{4 - x^2}}$

d. $\frac{\sqrt{4 - x^2}}{2}$

37 [Intermediate] • Target: Average rate of change from a graph

For $0 < x < 1$, simplify $\cos\left(\sin^{-1}\left(\frac{x}{3}\right)\right)$ as an algebraic expression in x.

a. $\frac{\sqrt{9 + x^2}}{3}$

b. $\sqrt{9 - x^2}$

c. $\frac{\sqrt{9 - x^2}}{3}$

d. $\frac{x}{\sqrt{9 - x^2}}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Units and interpretation in context • Cross-Lesson: 7.02 Logarithms and Their Properties

For $0 < x < 1$, simplify $\sin\left(\tan^{-1}\left(\frac{x}{4}\right)\right)$ as an algebraic expression in x.

a. $\frac{x}{\sqrt{x^2 + 16}}$

b. $\frac{x}{4}$

c. $\frac{x}{\sqrt{16 - x^2}}$

d. $\frac{4}{\sqrt{x^2 + 16}}$

39 [Challenging] • Target: Formula and Secant Line Concept • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

For $0 < x < 1$, simplify $\sin\left(\tan^{-1}\left(\frac{x}{3}\right)\right)$ as an algebraic expression in x.

a. $\frac{x}{\sqrt{9 - x^2}}$

b. $\frac{x}{3}$

c. $\frac{x}{\sqrt{x^2 + 9}}$

d. $\frac{3}{\sqrt{x^2 + 9}}$

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40 [Challenging] • Target: Tabular and Data-Based Rates of Change • Cross-Lesson: 7.04 Applications: Growth and Decay

For $0 < x < 1$, simplify $\sin\left(\cos^{-1}(x)\right)$ as an algebraic expression in x.

a. $\sqrt{1+x^2}$

b. $-\sqrt{1-x^2}$

c. $1-x$

d. $\sqrt{1-x^2}$

41 [Challenging] • Target: Algebraic Expressions with Variables • Cross-Lesson: 8.01 Rational Functions and Asymptotes

For $0 < x < 1$, simplify $\tan\left(\sin^{-1}(x)\right)$ as an algebraic expression in x.

a. $\frac{x}{\sqrt{1+x^2}}$

b. $\frac{x}{1-x^2}$

c. $\frac{x}{\sqrt{1-x^2}}$

d. $\frac{\sqrt{1-x^2}}{x}$

42 [Challenging] • Target: Contextual Applications • Cross-Lesson: 8.02 Graphing Rational Functions

For $0 < x < 1$, simplify $\tan\left(\cos^{-1}(x)\right)$ as an algebraic expression in x.

a. $\frac{\sqrt{1-x^2}}{x}$

b. $\frac{1-x^2}{x}$

c. $\frac{x}{\sqrt{1-x^2}}$

d. $\frac{\sqrt{1+x^2}}{x}$

43 [Challenging] • Target: Computing average rate of change • Cross-Lesson: 8.03 Introduction to Limits

For $0 < x < 1$, simplify $\sin\left(\tan^{-1}(x)\right)$ as an algebraic expression in x.

a. $\frac{x}{\sqrt{1+x^2}}$

b. $\frac{x}{1+x^2}$

c. $\frac{1}{\sqrt{1+x^2}}$

d. $\frac{x}{\sqrt{1-x^2}}$

44 [Challenging] • Target: Applications • Cross-Lesson: 9.01 Arithmetic Sequences and Series

For $0 < x < 1$, simplify $\cos\left(\tan^{-1}(x)\right)$ as an algebraic expression in x.

a. $\frac{1}{\sqrt{1+x^2}}$

b. $\sqrt{1+x^2}$

c. $\frac{1}{\sqrt{1-x^2}}$

d. $\frac{x}{\sqrt{1+x^2}}$

45 [Challenging] • Target: Comparing rates and estimating from a table • Cross-Lesson: 9.02 Geometric Sequences and Series

For $0 < x < 1$, simplify $\sin\left(2\cos^{-1}(x)\right)$ as an algebraic expression in x.

a. $2x\sqrt{1-x^2}$

b. $2x^2-1$

c. $2\sqrt{1-x^2}$

d. $x\sqrt{1-x^2}$

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